

Andrea Benedetti — bndndr.it

Data and services: designing in (un)desirable futures

Designing services beyond functionalities as spaces of conflicts and reinventions

Nice to meet you!

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@ Università degli Studi di Milano

Data and Service Design

Academic Year 2024-2025, 2nd semester



The course "Data and Service Design: Designing Data-centric Services" is part of the Master of Product Service System Design of the School of Design (A.Y. 2024/2025). It intends to provide service design students with the conceptual and theoretical tools to understand the "datafication of society" and integrate data as part of their service design practice.

Throughout an approach both theoretical and practical, the students will investigate the main topics in the contemporary data landscape. They will relate to them the service design perspective under the notion of data-centric services, while engaging in data practices hands-on.

The learning path will follow three stages: 1) Data-driven research for service design; 2) Data-driven representations for service design; and 3) Designing data-centric services.

The course aims to provide service design students with the conceptual and practical tools for critically understanding the role of data in society, relate it to their service design perspective, and make data an integral part of their service design practice.

Students are invited to engage with the concept of "data" by following three guiding principles:

1. Everything could potentially be seen as data
2. Data is not a neutral material
3. Data supports both "research for" and "ideation of" new services

Starting from these premises, the class will approach the course topic through a three-stage learning framework. Each stage will include dedicated lectures, hands-on activities and involve specific guest experts.

Private and public organizations harness vast quantities of data to optimize their services. The adoption of data-driven services has become prominent across multiple sectors, including welfare, healthcare, taxation, and mobility.

These services often incorporate semi-automated systems that leverage data analytics and user profiling to enhance performance aspects within the supporting digital infrastructure. In the past years, some of them, hindered by biases in data and flawed design, have led to harmful outcomes. Notable examples include the scandals connected to Robovoxit Scheme in Australia, childcare benefits in the Netherlands, the surge in Uber prices as people fled Hurricane Katrina, or racist labels due to biased dataset in content recognition algorithms. These cases clearly illustrate how today services stand at the intersection between digital automation, organizations' strategies, users' interaction, regulations and ethics.

This centrality demands that service designers develop their literacy about data and related socio-technical dimensions. In their designing, services designers should consider how data interplays with value co-creation — within, and beyond, the services systems they design for.

Course perspective

From Data-Driven Services to Designing for Data-Centric Services

Data practices for service designers

Data collection and storage have become routine for public and private organizations. In particular, administrative and service data (also called micro-data), are routinely generated by private and public data holders throughout their activities and services. Consequently, several organizations find themselves in the position of being data holders, in other words, as a by-product of their activities, they possess relevant data that can be turned into value. However, to effectively utilize this data as a tool as addressing significant interoperability barriers — including accessibility, management, and representation — while ensuring compliance with legal and technical frameworks.

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Against this background, the course advances the concept of Data-Centric Services. We emphasize the importance of designing for services by leveraging data generated by scattered digital information systems — not yet linked, connected, or used coherently. Service designers can engage with data-centric services as they bring a service perspective useful for making sense and creating value of scattered data sources, while offering data holders sense-making and building collaboration among them.

With data-centric services, Data Practices represent it is course's after certain concept, that is particularly necessary to acknowledge data's cross-sectoral and biased nature. Standing in contrast to an abstract idea of data, the notion of data practice reminds students are scholars to recognize it as a material that is deeply made by someone (Burkhardt, et al., 2022).

Data is, therefore, intrinsically linked to social, political, organizational and historic conditions. The pervasiveness of data in all fields — and the strong relationship between the two — require future service designers to engage with data as one of the materials of their practices, acknowledging not only the operational attributes of data in services, but also economic and ethical ones (Dell, 2020). This includes considerations such as the public acceptability of data collection, appropriate means to make use of all available data, and what will be the ultimate effects of data, both in service systems' key design for and beyond them.

Through these two topics, this course asks students to promote the provenance of data at the service research and prototyping level, critically engage in data collection, representation, and visualization, and develop insights from data and to involve stakeholders throughout their service design activity.

Course structure

Data-driven research for service design

This stage of the course will focus on the topics of accessing and mobilizing heterogeneous data resources, such as those available from the web and social media. At this stage, students will be invited not only to learn how to interact with data but also to understand "how" or "how" datasets have been constructed and for which representation purposes.

Data-driven service design representation

Service designers must service representations and conveying knowledge with final users and stakeholders (e.g., to refine initial service ideas). This stage of the course will focus on how to work with data in service representation perspective, and which methods/tools can be used for the collection, analysis and visualization of heterogeneous data sources.

Designing data-centric services

Service designers who seek to develop new services centered on available data must engage with dimensions pertaining to regulation, privacy, data governance and ownership. Moreover, they must consider the legal and ethical implications of accessing and using data. But they also need to be able to use, at this stage, the students will be asked to self-reflect on the data they worked with in the previous stage and imagine possible ideas of business centered on the connection between external data sources.

Guest lecturers



February 26, 2025

Paolo Giacomelli - Northwestern University, Politecnico di Milano

Title: "Design of Data Perspectives"



February 27, 2025

Yarel A. Loukianov - Georgia Tech Institute

Title: "Open Data Settings"



March 6, 2025

Bulk Lund - Samenoms/afzondering Centrale Organisatie

Title: "Evaluating the impact of Design and experience as a model guide for developing Regional digital services based on social data and user research"



March 13, 2025

Paola Trilli - PagineGialle

Title: "Data-driven service design in public"



March 20, 2025

Gabriele Colombo - Politecnico di Milano

Title: "The web as a source of data and production of digital methods"



March 23, 2025

Roberto Belmonte - Better Public Services

Title: "The business and governance of citizens' data in the cycle of public digital services"



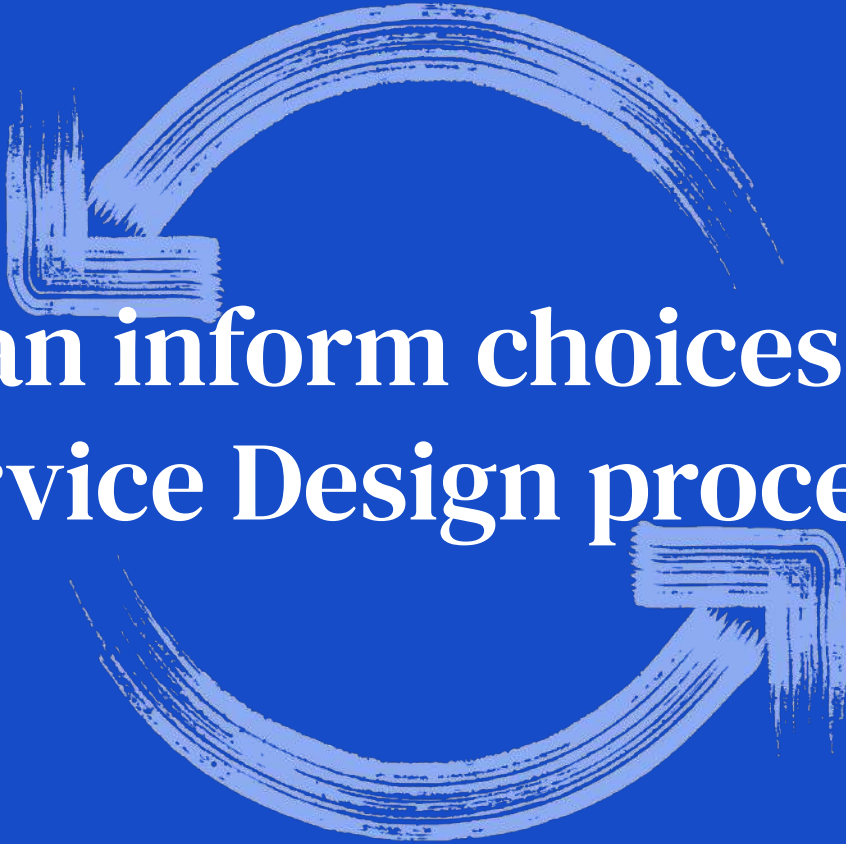
April 10, 2025

Petter Falk - Karlstad University

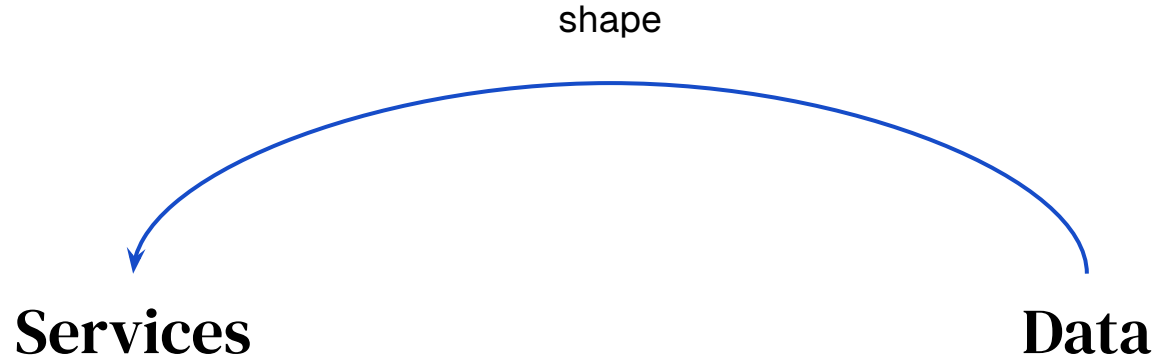
Title: "Data-driven democracy: From governance to value in a data-driven public services"

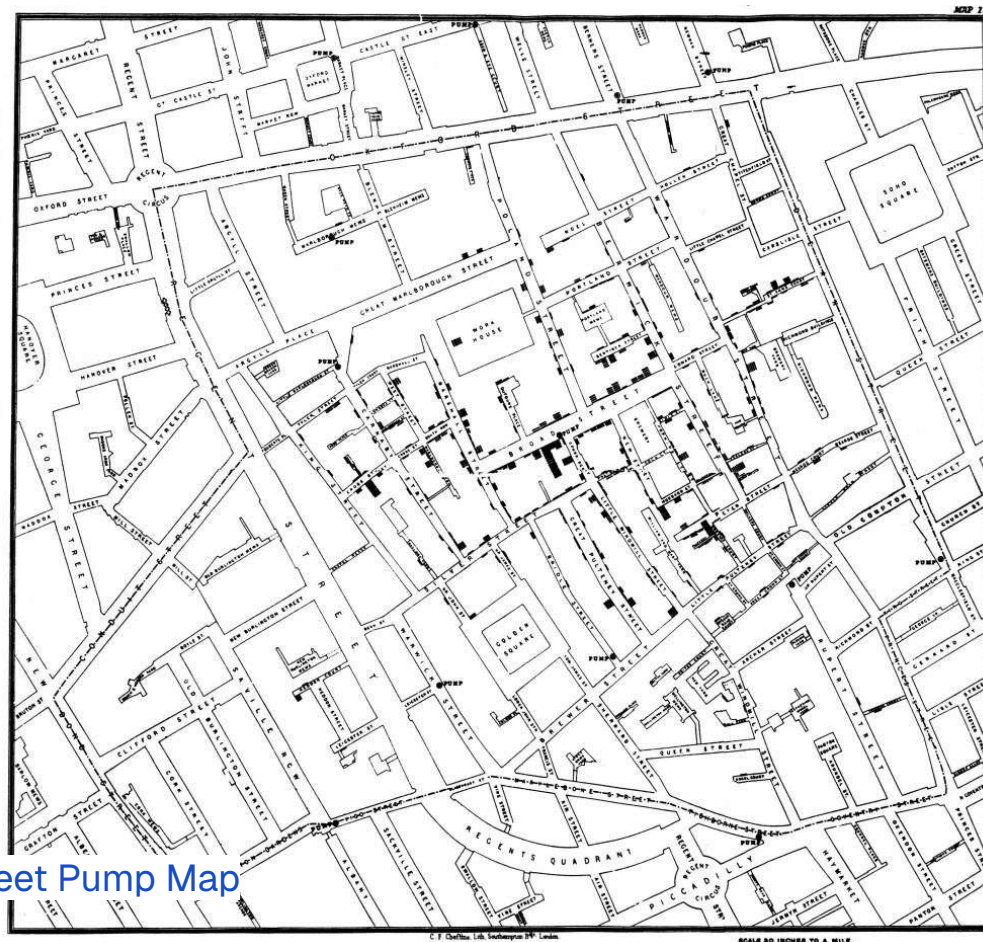
What we learned last year

1. Data can inform choices in the Service Design process
2. Data present ethical and ontological questions to service designers
3. Data is a core infrastructure of private and public services

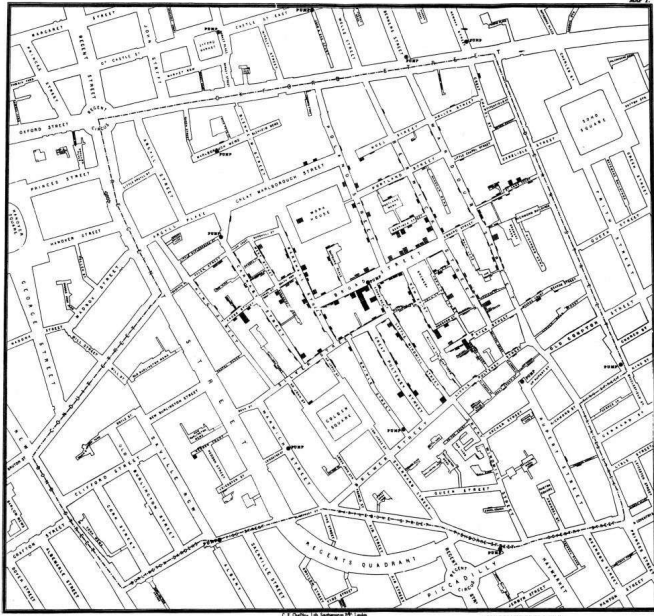


Data can inform choices in the Service Design process





John Snow's Broad Street Pump Map



Hypothesis

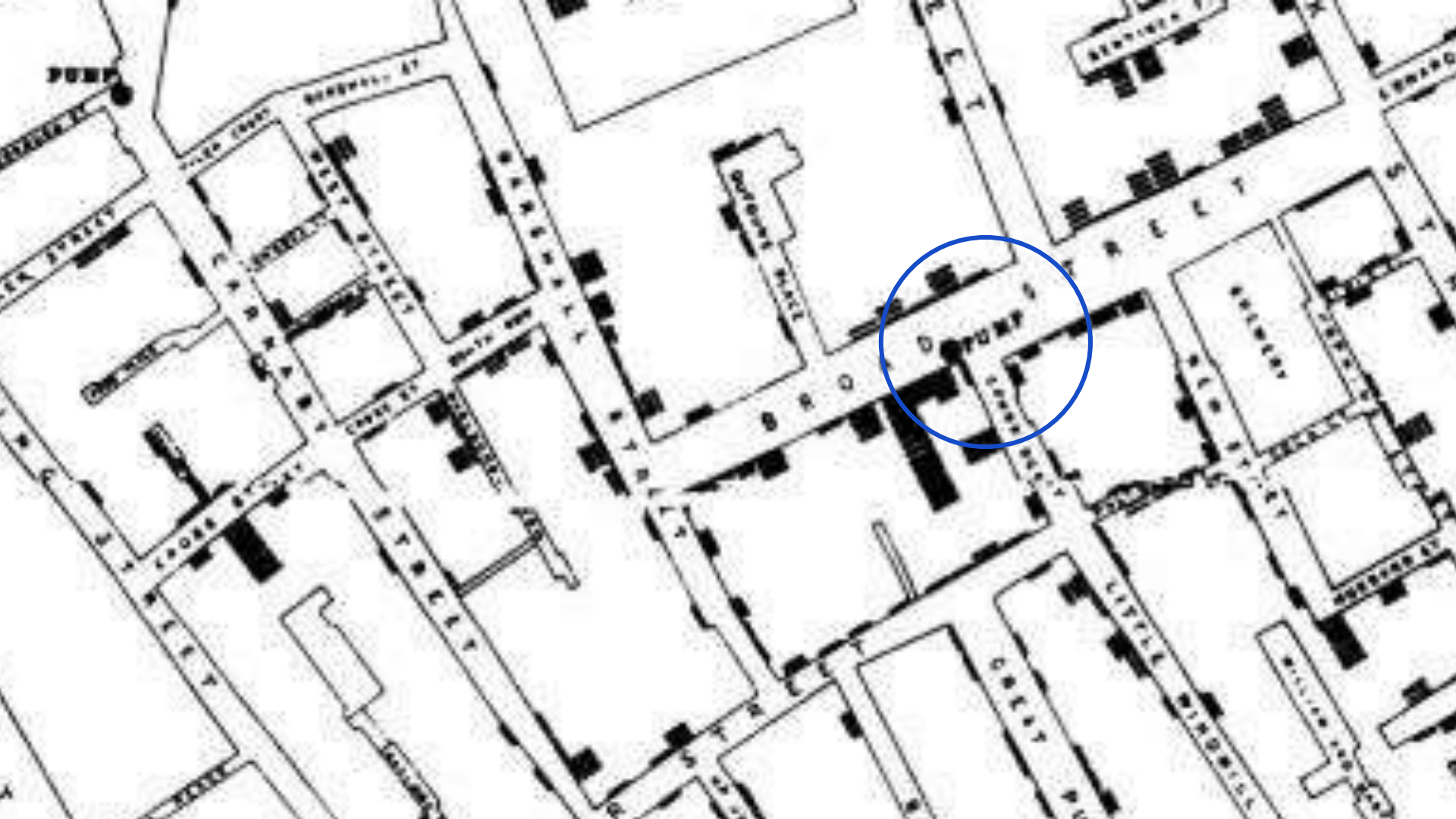
Cholera spreads through water and not air

Data collection

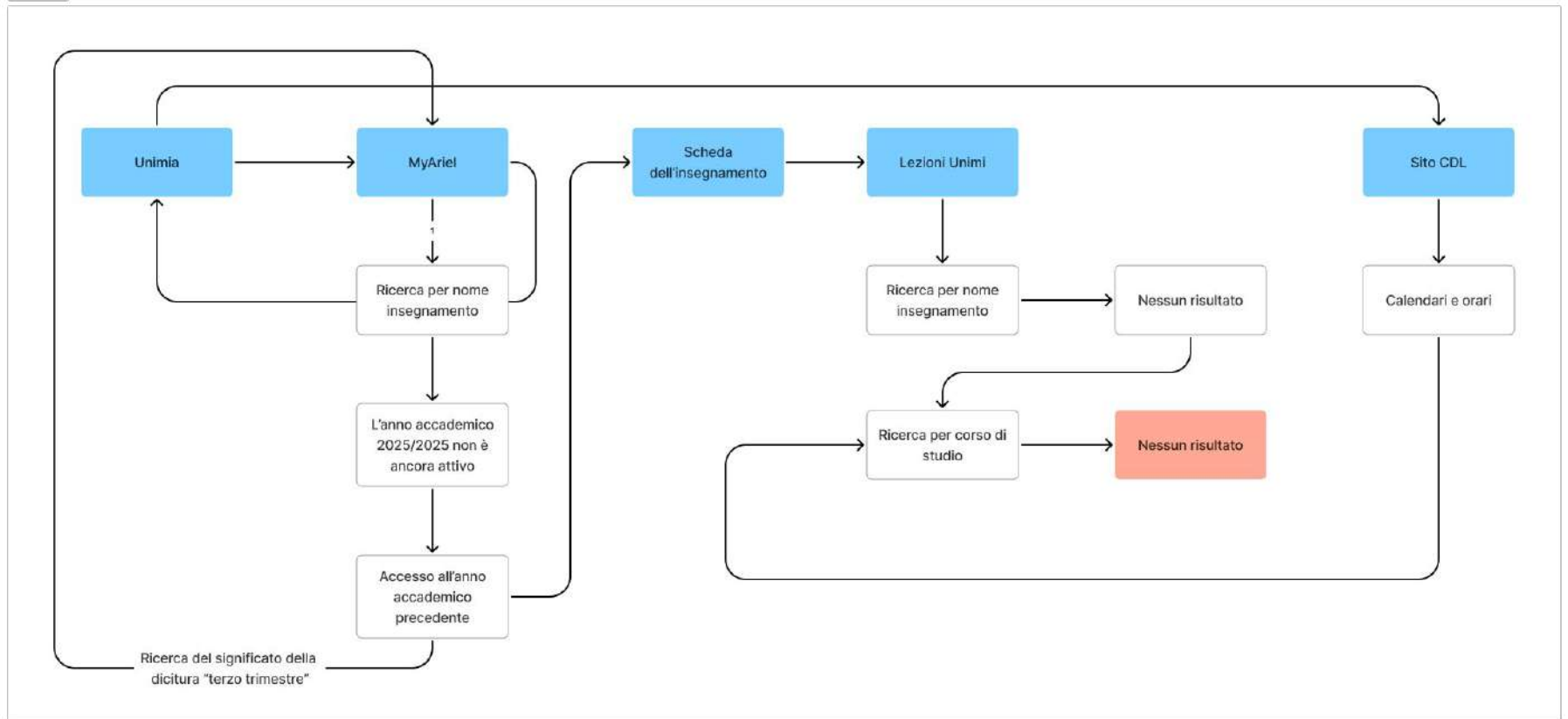
Location and number of deaths

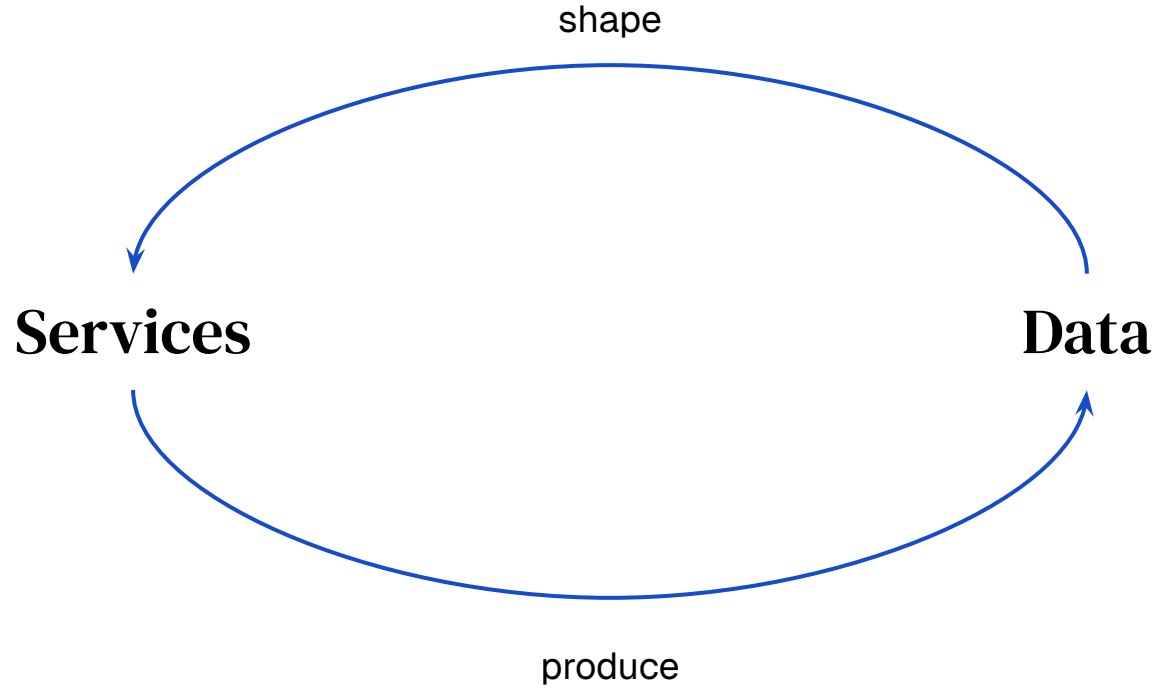
Visualization

Compared to points of interest in the neighborhood, where do people die the most of cholera?



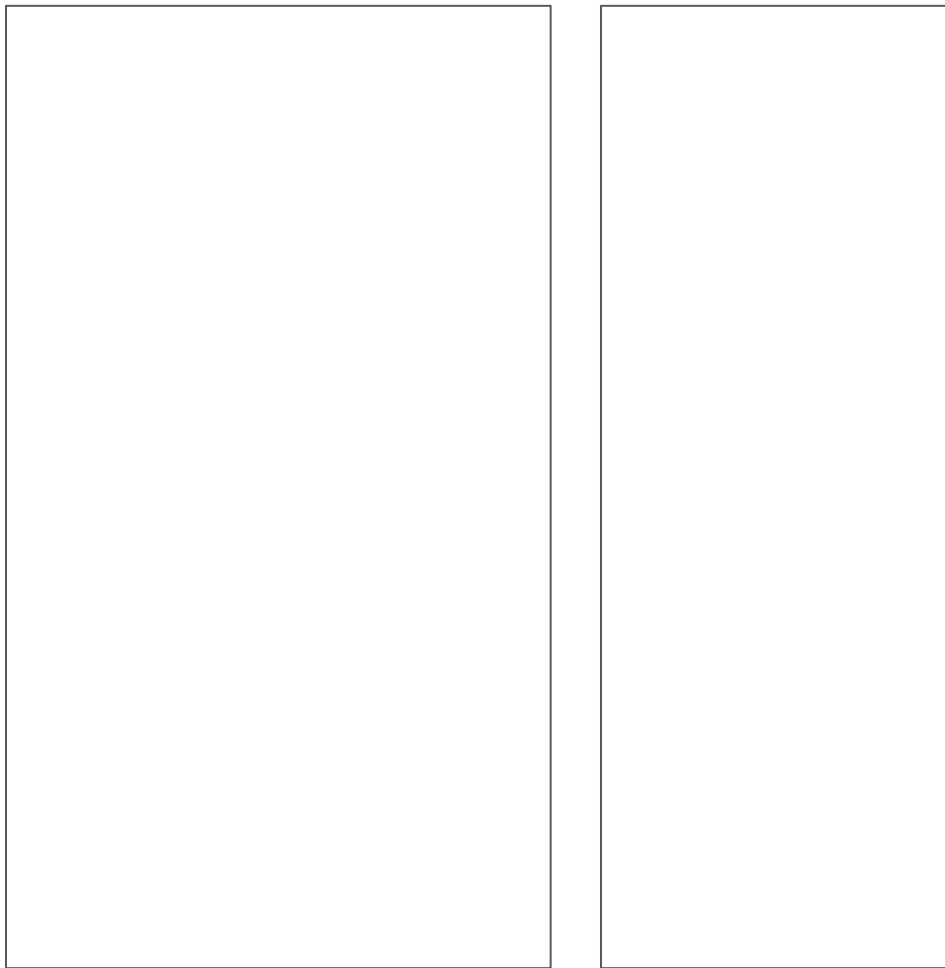






“Digital mediation adds to collective phenomena a couple of precious properties: *traceability and aggregability.*”

Venturini, 2012



Costituito il Comitato unico di garanzia per il triennio 2026-2028

Leggi ▶

Selezione per un incarico post-doc >

Milano Cortina 2026: chiusura
sedi il 6 febbraio >

Selezione per un incarico post-doc >

Selezioni per 5 contrattisti di
ricerca >

Selezione per 3 contrattisti di
ricerca >

Tutti gli avvisi ▶

? Cosa fare se

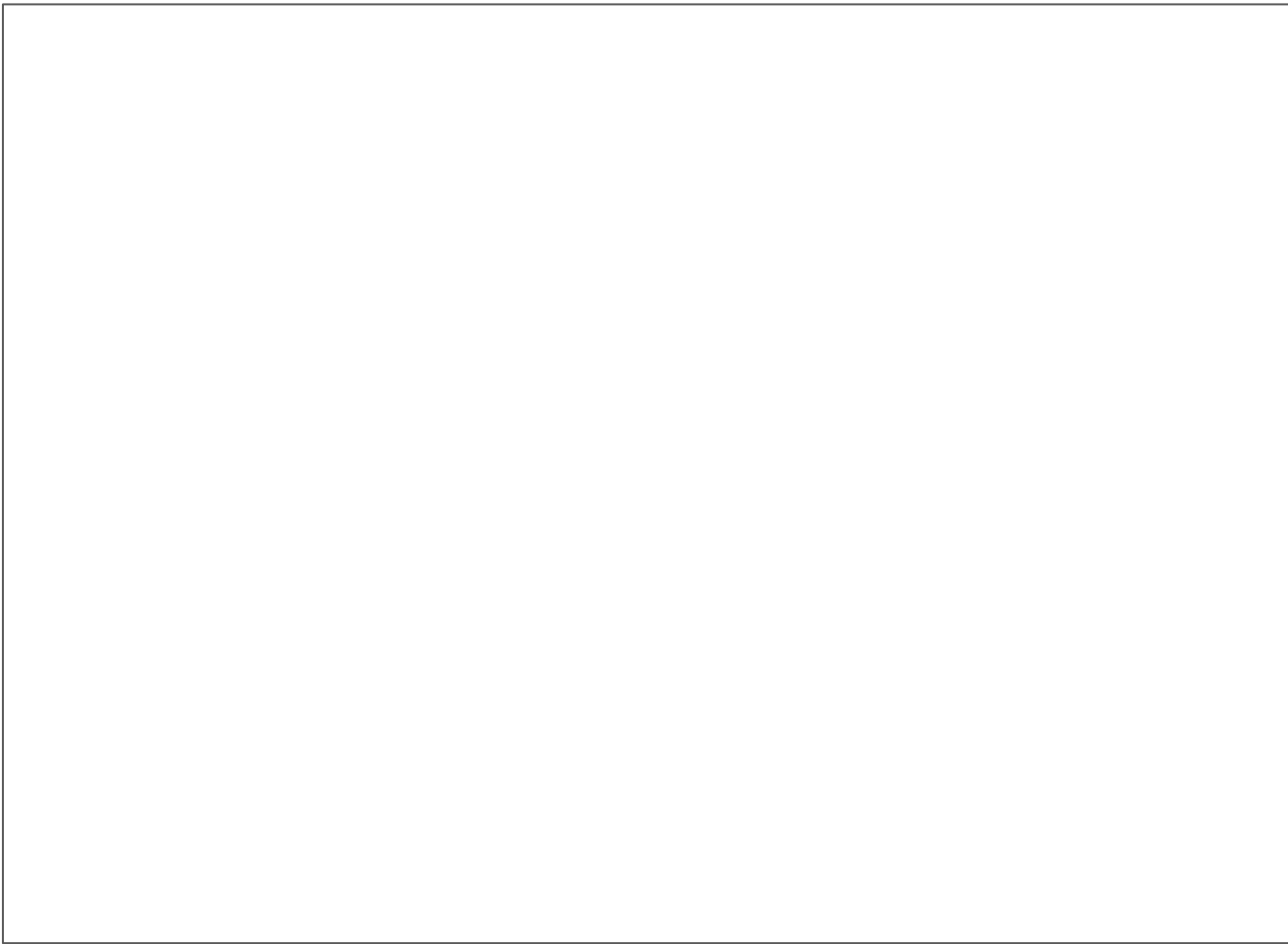
Ho bisogno del WiFi

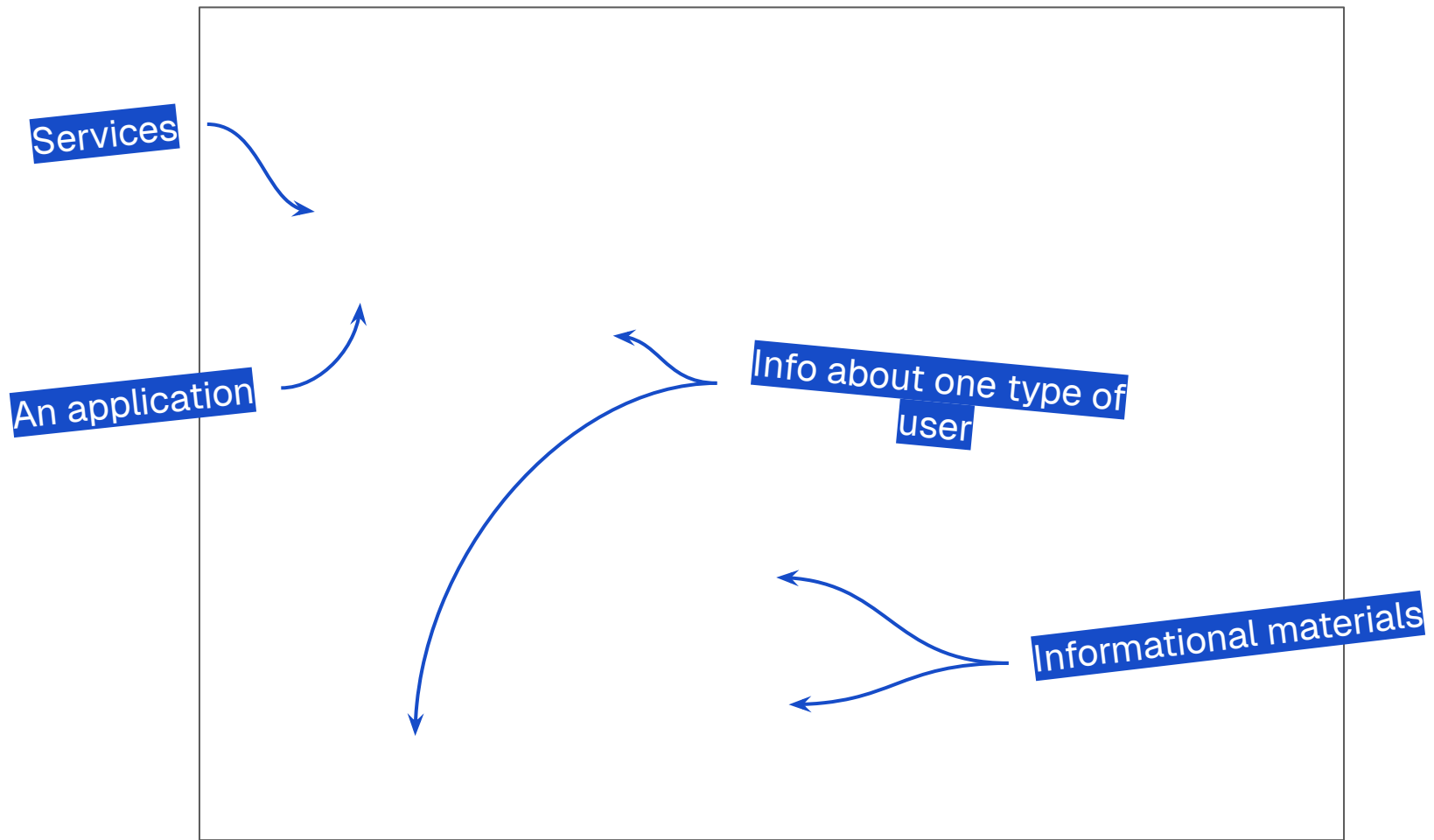
! I più richiesti

Welfare , famiglia e agevolazioni

➔ Accesso rapido

Presenze e giustificativi







Data presents ethical and ontological questions to service designers

Data are capta

interpretative framework on which the statistical data were constructed. The graphical force conceals what the statistician knows very well—that no “data” pre-exist their parameterization. *Data are capta*, taken not given, constructed as an interpretation of the phenomenal world, not inherent in it.

To expose the constructedness of data as capta a number of systematic changes have to be applied to the creation of graphical displays. That is the foundation and purpose of a *humanistic approach* to the qualitative display of graphical

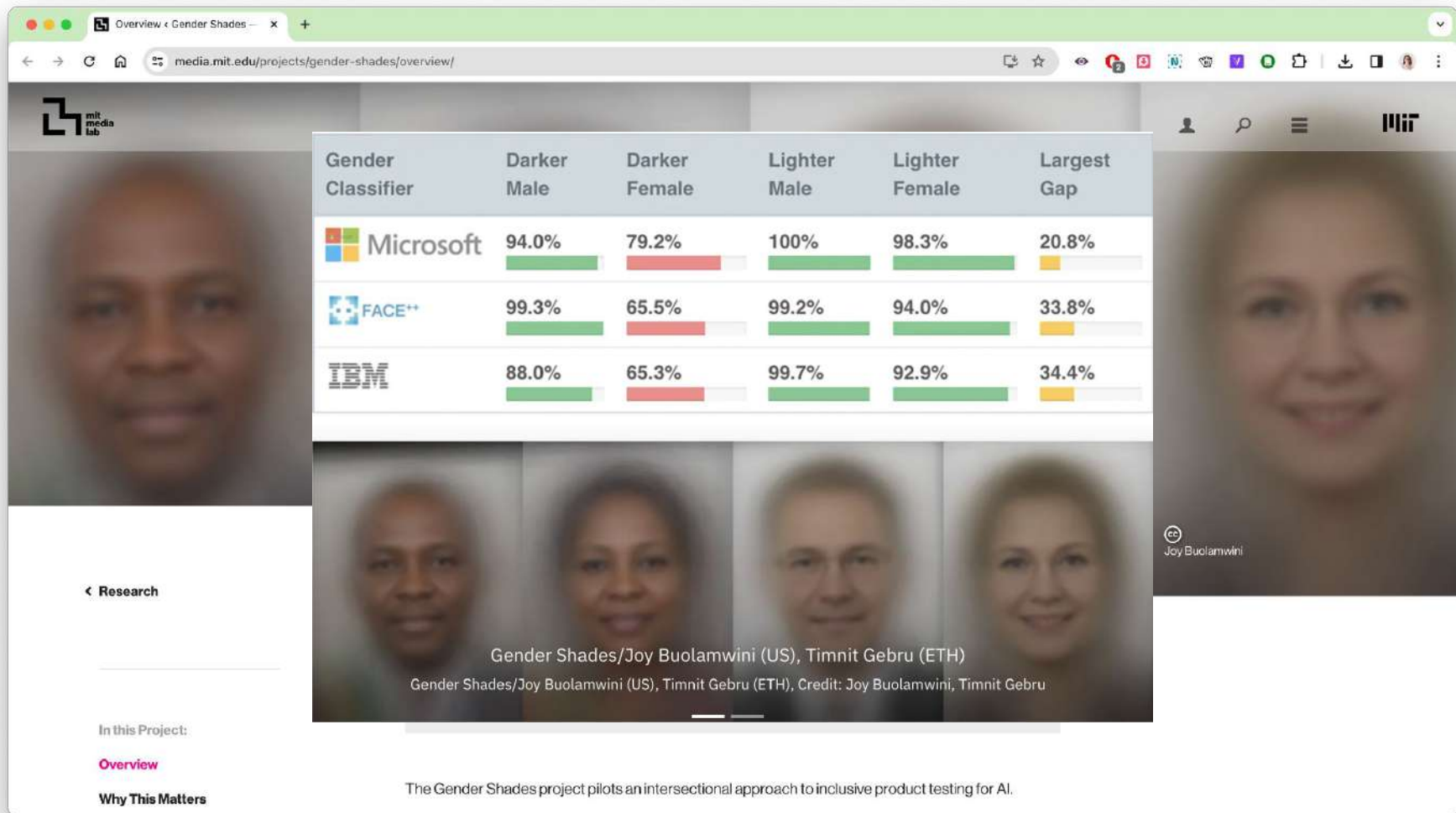
Johanna Drucker, 2011

The animals are divided into: (a) belonging to the emperor, (b) embalmed, (c) tame, (d) sucking pigs, (e) sirens, (f) fabulous, (g) stray dogs, (h) included in the present classification, (i) frenzied, (j) innumerable, (k) drawn with a very fine camelhair brush, (l) et cetera, (m) having just broken the water pitcher, (n) that from a long way off look like flies.

J.L. Borges. The Analytical Language of John Wilkins

Technologies are crystallizations of society: they bear the imprint of the people and social context in which they develop.

Wajcman, J. (2019). How Silicon Valley sets time. *New Media & Society*, 21(6), 1272–1289. <https://doi.org/10.1177/1461444818820073>





Petter Falk, Karlstad University

User data in the private sector
generates money, but what is *data* in
the public sector?

Petter Falk, Data and Service Design A.A. 2024/2025



Data is a core infrastructure of
private and public services

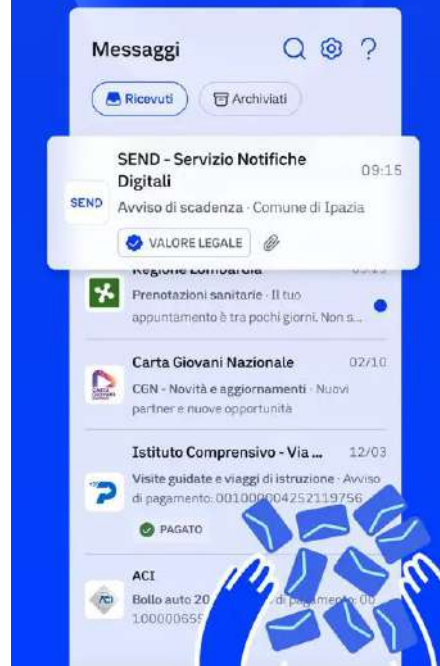


Agata Brilli, PagoPA

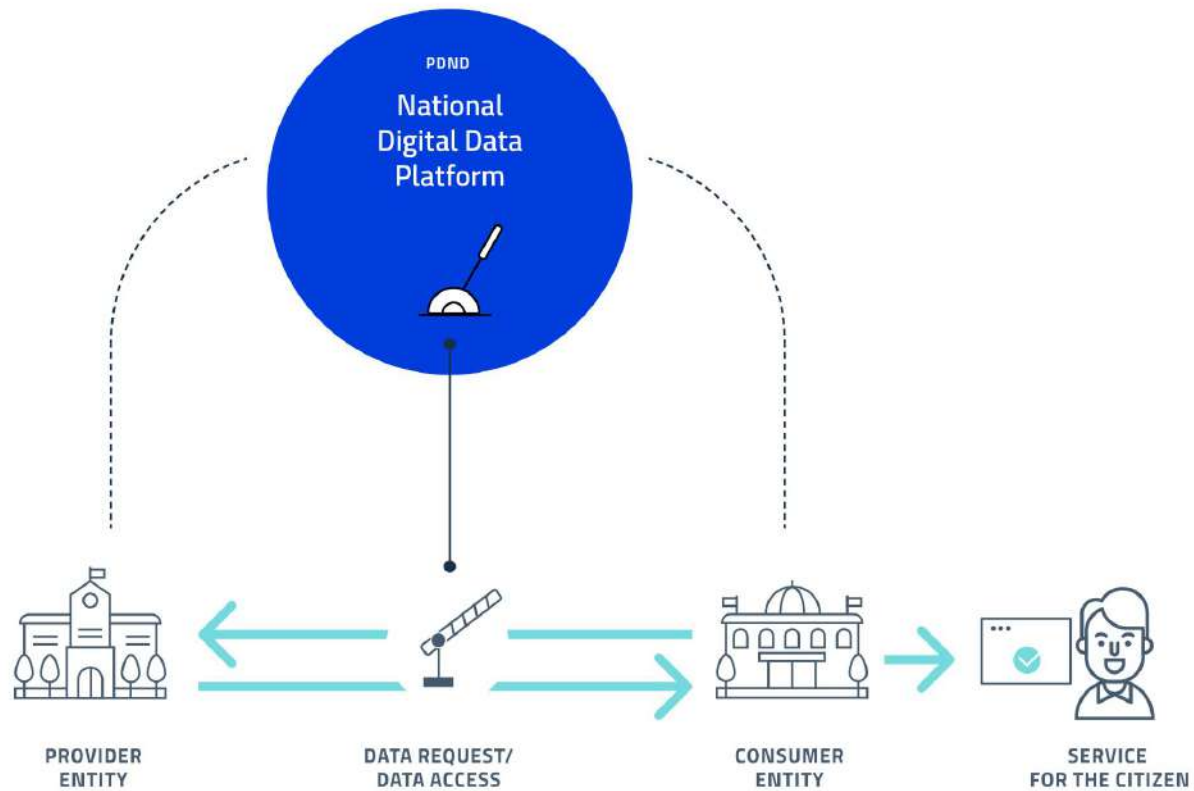
Conserva al sicuro
i tuoi documenti
in versione digitale



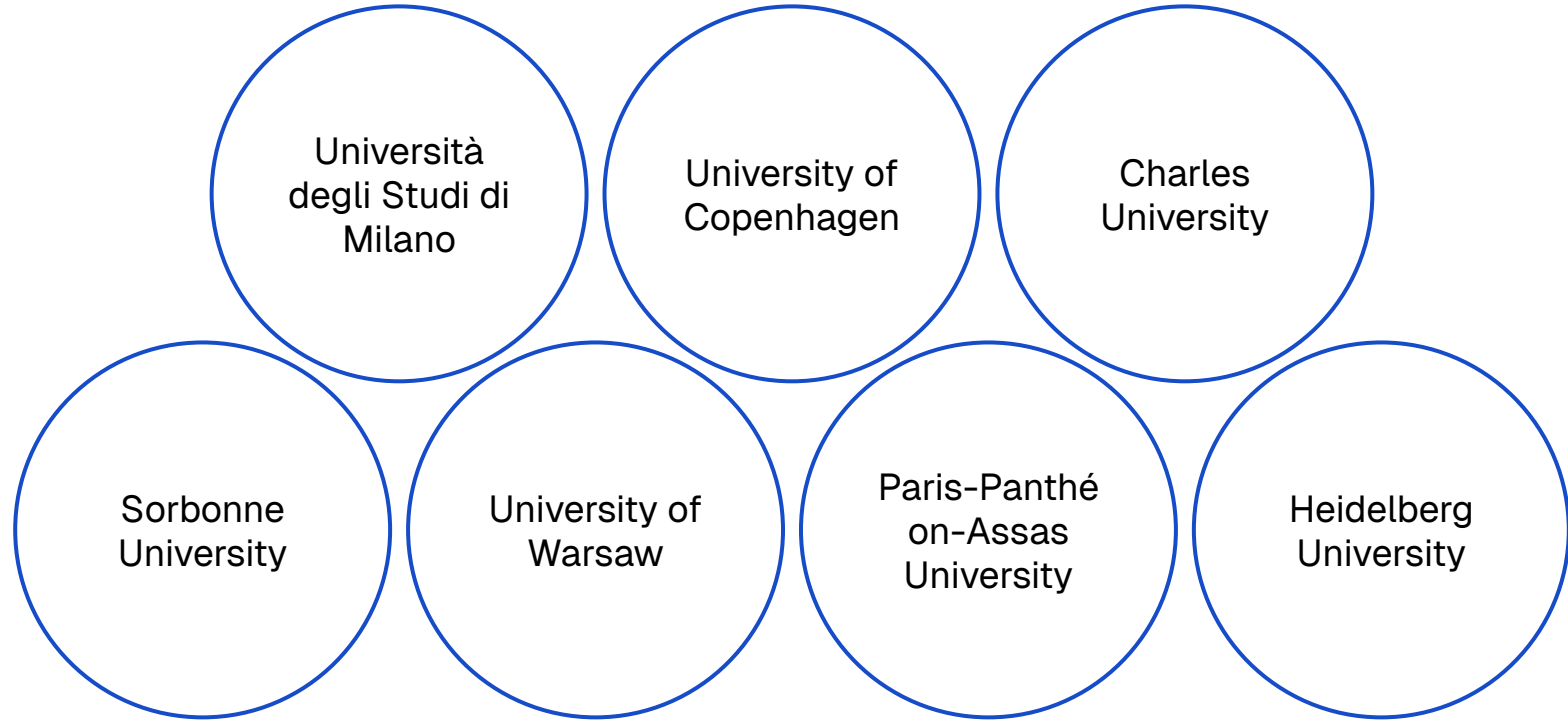
Ricevi messaggi
e comunicazioni,
anche a **valore legale**

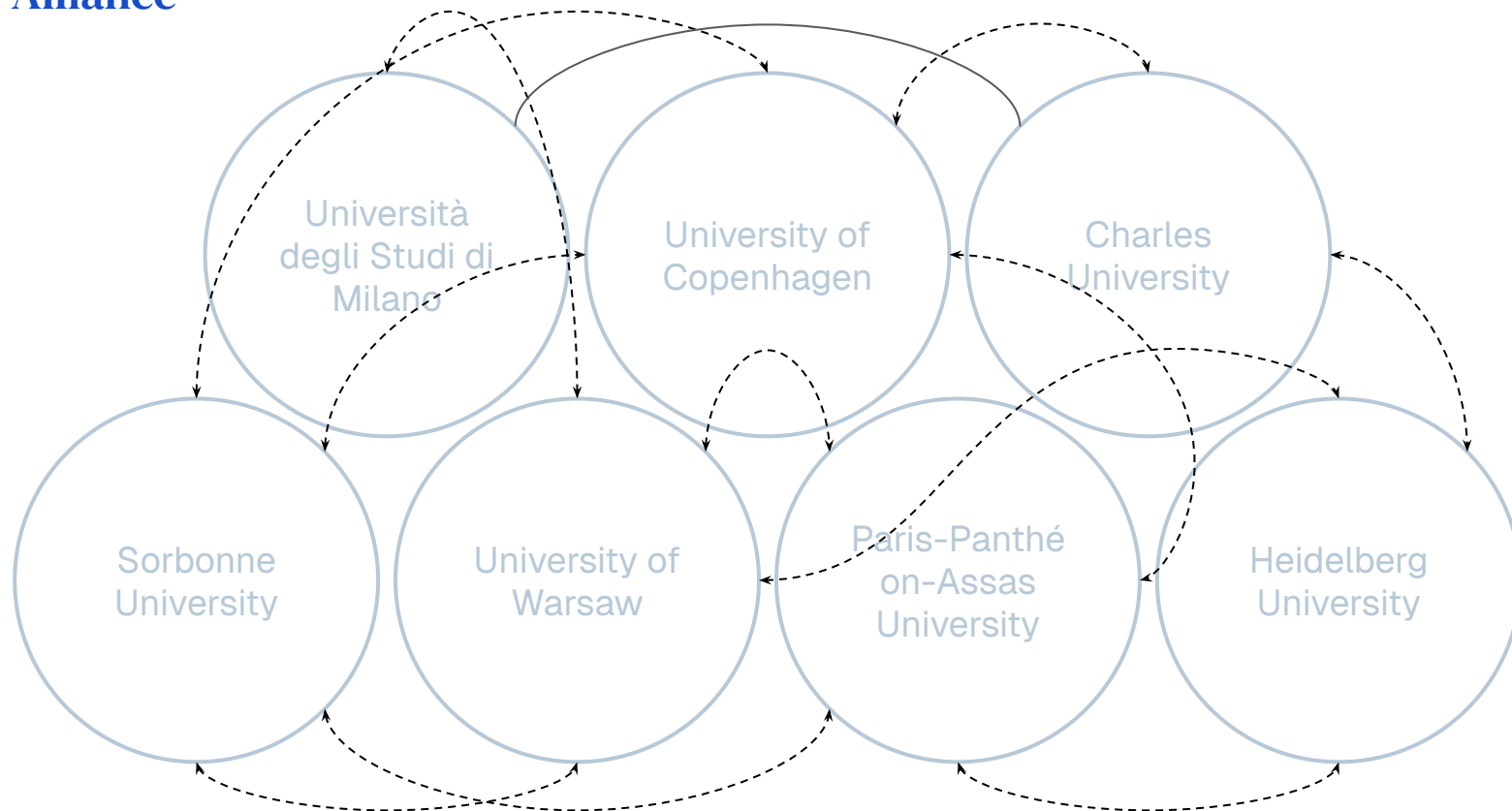


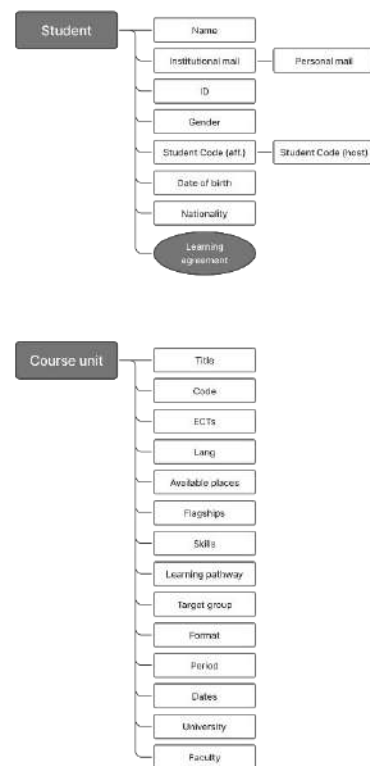
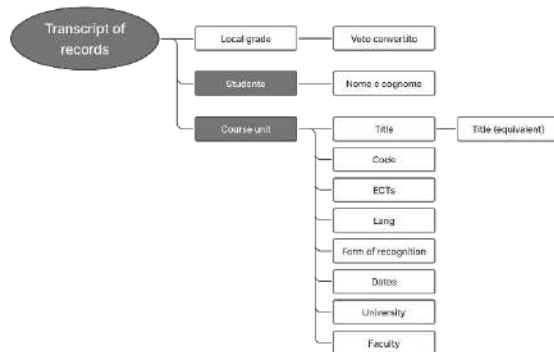
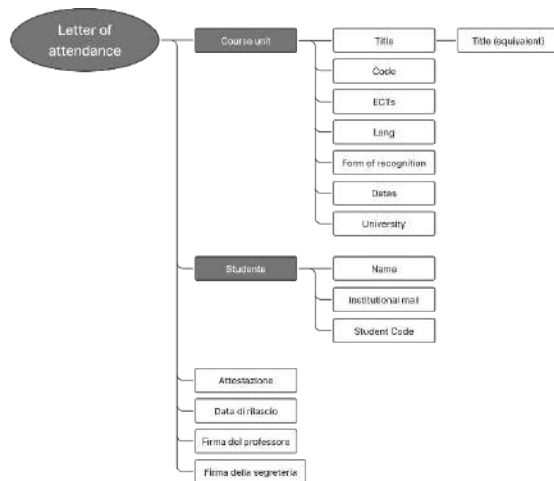
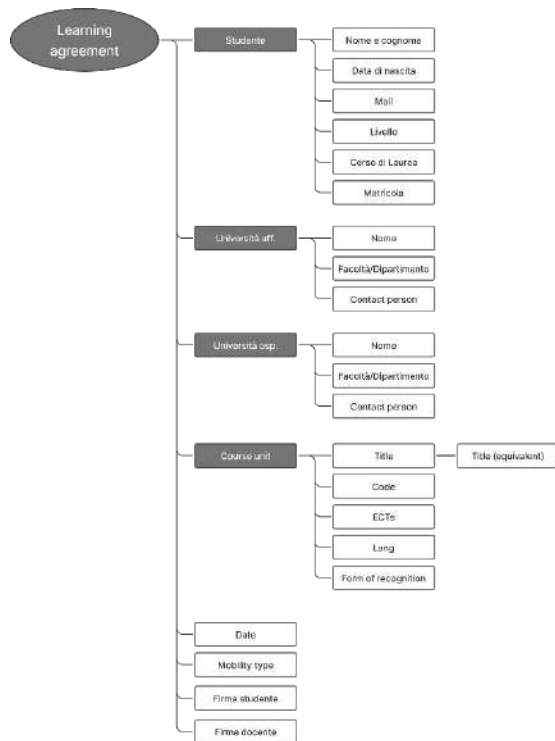
Interoperability



Messiness



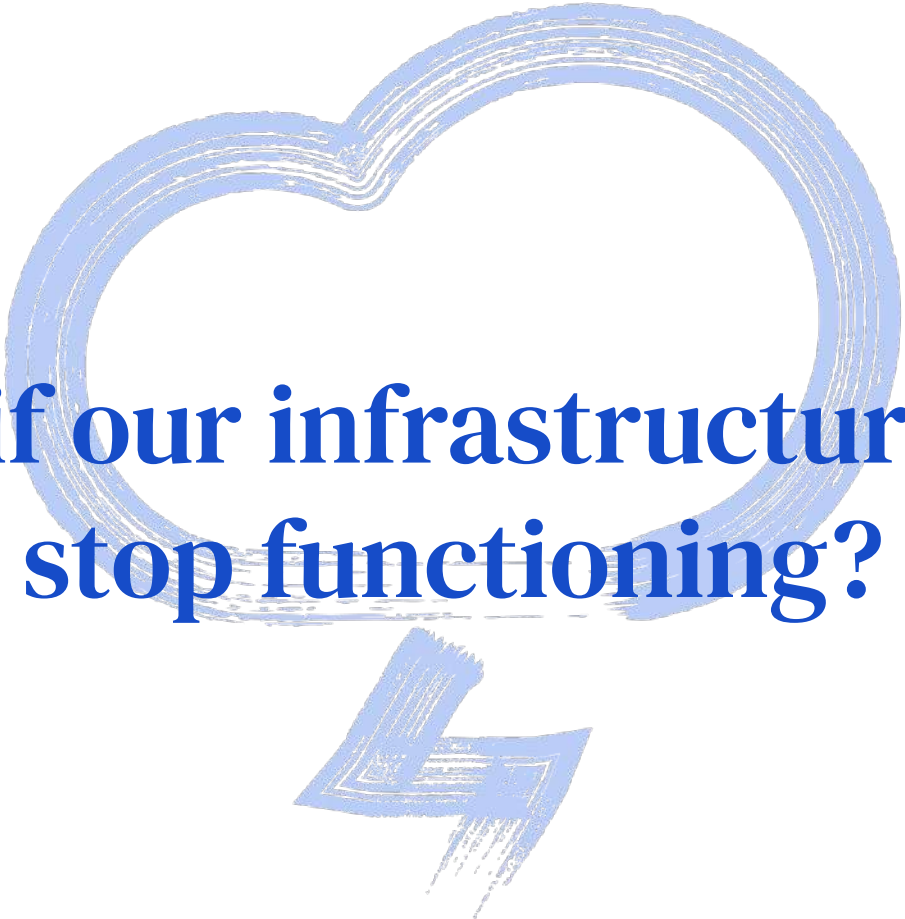




Messiness comes from

1. Misalignments in technological infrastructures
2. Technical incompatibilities
3. Legal incompatibilities
4. Plain refusal
5. ...

Are we living in (un)desirable futures?



**What if our infrastructures will
stop functioning?**



Amazon.it · Disponi...
Introduction To Ch...



Amazon UK
Supercharge Your ...



ManageEngine
All about AI workflows | AppCreator



Insidr.ai
The 31+ Best AI Productivity Tools to S...



Goodreads
Introduction To Ch...



LinkedIn
AI tools to supercharge your productivity



YouTube
Supercharge Your Productivity with AI Tools fo...



tCognition
AI Tools to Supercharge Your Work



YouTube
Using generative AI to supercharge your prod...



LinkedIn
Supercharge Your Productivity and Career with ...



X
HyperGPT on X: "How AI Can Superch...



Medium
Top 10 Mind-Blowing AI Tools to Super...



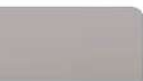
Medium
Supercharge Your Productivity: 10 Mus...



YouTube
Supercharge Your Productivity with Ch...



Oneboard
How AI Can Supercharge Your...

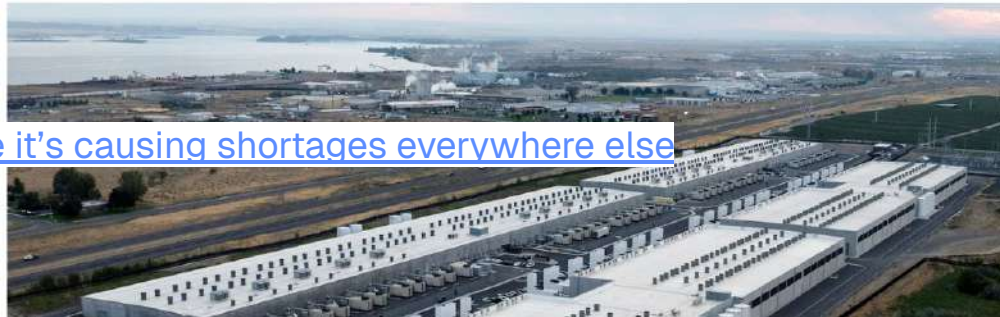


The AI boom is so huge it's causing shortages everywhere else

The \$700 billion AI spending spree has few precedents. Good luck finding an electrician or a reasonably priced smartphone.

February 7, 2026

6 min 1,464



[The AI boom is so huge it's causing shortages everywhere else](#)

Expect HDD, SSD shortages as AI rewrites the rules of storage hierarchy — multiple companies announce price hikes, too

News By Zak Killian published September 15, 2025

Computing

RAM prices keep going up — what is RAMageddon, and why is it getting worse?

News By Jason England published 2 days ago

Pretty much everything tech-wise is affected

BIG TECH • CHIPS

Rampant AI demand for memory is fueling a growing chip crisis

By Debby Wu, Takashi Mochizuki, Yoolim Lee, and Bloomberg

February 15, 2026, 7:42 PM ET

[Add us on](#)



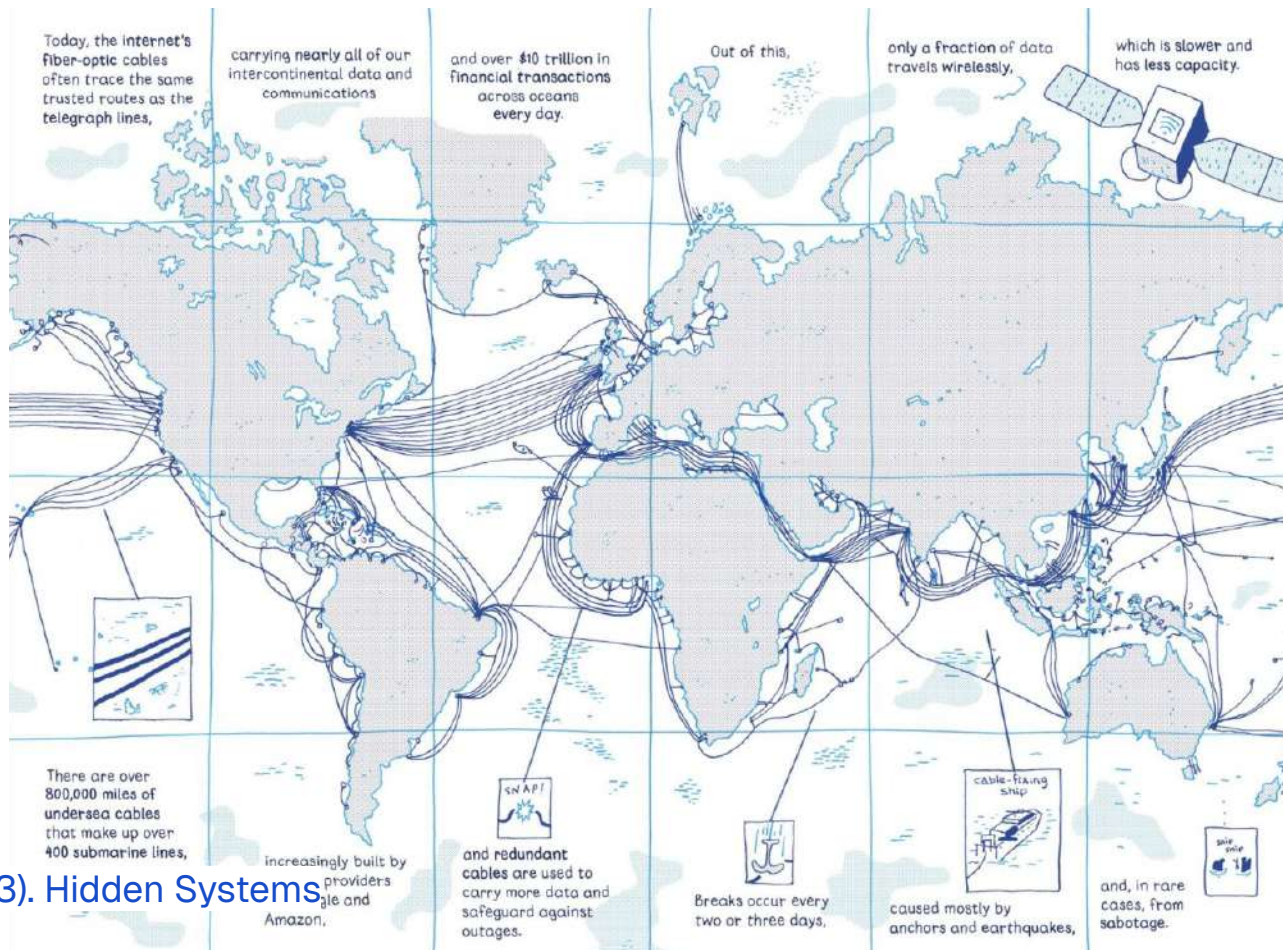
“By directly harnessing near-constant solar power with little operating or maintenance cost, these **satellites will achieve **transformative cost and energy efficiency** while significantly reducing the environmental impact associated with terrestrial data centers”**

Elon Musk's Orbital Data Centers Face Huge Challenges

“The cost of transporting all the building materials and components into orbit followed by constructing the platform in space itself to form the data center would be **quite expensive**, even if it’s built in modules”

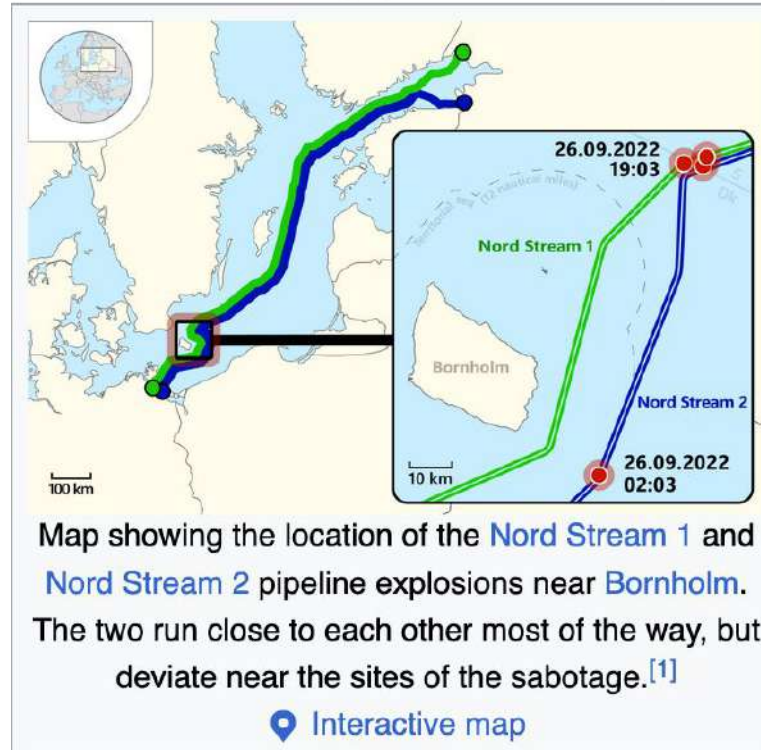
ibid.





Dan Nott (2023). Hidden Systems

Nord Stream pipelines sabotage



[Nord Stream pipelines sabotage - Wikipedia](#)

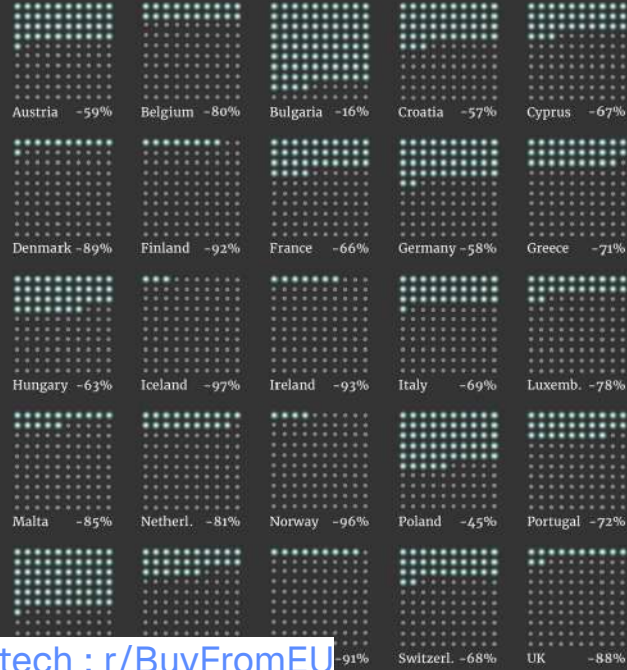


What if services are weaponized to cause harm?

If Trump flips the switch

Without access to U.S. tech, over 74% of publicly listed European companies would go dark

Bulgaria would be the least affected, with 84% of companies being able to resume their activities if Trump pulls the plug on Europe.

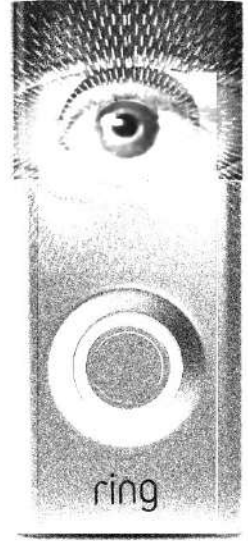


Source: Proton (2025)

[Which countries rely most on US tech : r/BuyFromEU](https://r/BuyFromEU)



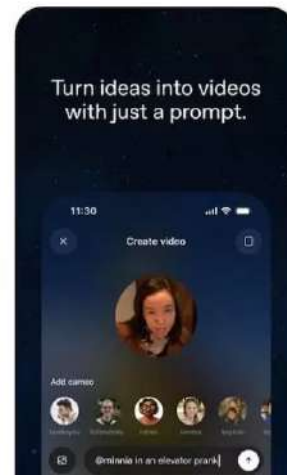
“This definitely isn’t about dogs — it’s about mass surveillance.”

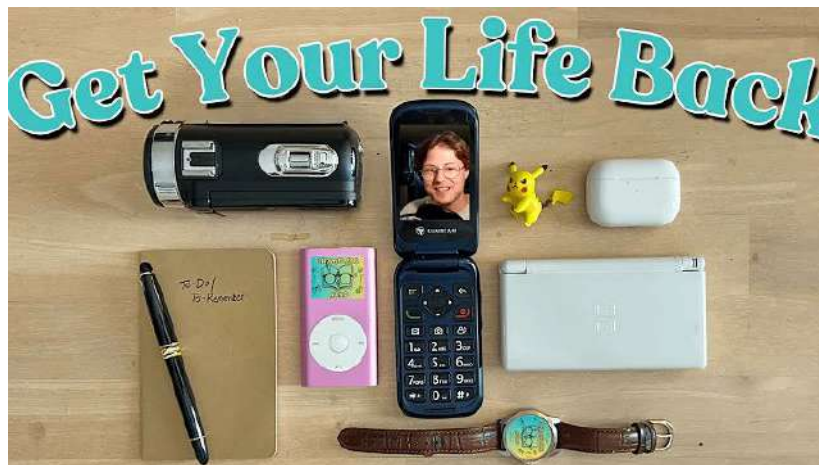


[Let's talk about Ring, lost dogs, and the surveillance state | The Verge](#)



**What if people start to reject new
services or technologies?**





“If these are three types of uncomputability — analog life, rational paradox, and practical limit — a fourth meaning of the term “uncomputable” derives from the [...] chaos, contingency, indefiniteness, change [...] it exists in the cracks and excesses of data.”

Uncomputable: Play and Politics In the Long Digital Age, Alexander R. Galloway

People may not want to participate



Curiosity  Colonization

Il designer senza qualità (2025), Silvio Lorusso

“Designers are respectable, curious, generous, [...] designers invade without colonizing.”

Paola Antonelli

“Design is a product of colonization.”

Ruha Benjamin

**Design services that can be
stepped out from, inquired,
broken up, tore down, dismantled,
rebuilt from scratch,
reappropriated, recontextualized,
made accountable...**

When looking at a dataset

1. What **type of data** is it? Qualitative, quantitative...?
2. Are there **empty spaces** in the column? Is the entire column consistent or are different ways of conveying information?
3. **Who** collected the data?
4. **How** did they collect the data? With which instruments?
5. **Why** did they collect the data?
6. **For who** is the data collected?

Data feminism

(D'Ignazio & Klein, 2020)

Data justice

Data activism

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The end

Thank you for listening!